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## PROJECTION-TRANSPORT STEERING: CONDITIONAL CONTROL OF REASONING BEHAVIOR IN LLMS

**Abstract.** Activation steering can change language-model behavior, but global residual-stream shifts may suppress useful and harmful behavior together. For reasoning confidence, the standard additive edit removes overconfident errors only by collapsing justified confidence, so apparent calibration gains can reflect indiscriminate hedging rather than better control. We introduce Projection-Transport Steering, which treats steering as a projection-dependent transport problem and applies an action only when a separate trace-level detector fires. This separates the decision to intervene from the intervention itself, allowing abnormal activation tails to move while typical states remain largely untouched. Across five resampled MMLU subsets, gated ablation reaches selectivity  $+0.266$  with no detectable accuracy change; additive steering provides essentially no selectivity and lowers accuracy. In a held-out OLMo-2/TruthfulQA evaluation, the detector-axis action improves calibrated truthfulness and correct-answer selection over a norm-matched action, with all seven predeclared construction controls passing. Formal results characterize the minimum projection-local intervention cost and detector-score separability under stated assumptions. These findings show that selective behavior control depends less on pushing activations harder than on separating detection from action. Steering systems should first determine when intervention is warranted, then use an input-dependent, projection-local edit.

### §1. Introduction

Inference-time activation steering promises behavior control without retraining: add a direction to the residual stream and the model refuses less, hedges more, or changes style. The workhorse is contrastive activation addition (CAA) [13]: a fixed vector, a hand-tuned coefficient, applied to every token. Yet applied to a concrete reasoning behavior; overconfidence of gemma-2-2b-it on multiple-choice reasoning; CAA fails in a way no

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coefficient tuning fixes: the dose that eliminates overconfident-wrong answers (OCW  $52 \rightarrow 0$  under a calibrated readout) simultaneously eliminates confident-right ones (CR  $77 \rightarrow 1$ ) and costs 8.3 accuracy points.

This paper treats that failure as a mathematical object rather than an engineering nuisance. Our contributions:

- (1) A unifying framework with optimality theorems. Every linear steering operator is a scalar transfer function  $g$  on the projection  $p = \langle h, \hat{v} \rangle$  (additive:  $g(p)=p+\alpha$ ; ablation:  $g=0$ ; CAST-style gating [8]:  $g(p)=p+\alpha\mathbf{1}[p>\tau]$ ). The optimal-transport map between projection laws is the unique minimal-perturbation intervention; additive steering is optimal only for pure location shifts (Thm. 1, Cor. 1).
- (2) A quantified impossibility result and its repair. Selectivity of any transfer function on an axis is capped by subpopulation separability along it (Thm. 2b); we measure token-level  $d = 0.049$  between OCW and CR on the confidence axis; the additive collapse is geometric necessity. The repair is conditional: an LDA gate on a second, detection axis is Neyman–Pearson optimal, and gate $\times$ transport is jointly optimal in its class (Thm. 3).
- (3) Strong trade-offs at minimal action cost in the evaluated setting. Across five MMLU seeds our conditional operators reach selectivity  $\approx +0.2$ – $0.27$  (additive  $\approx 0$ , prompting  $+0.07$ ) at zero accuracy cost, dominate faithful reimplementations of CAST, MiMiC, and Linear-AcT on the selectivity–accuracy–compute frontier, and replicate at  $4.5\times$  scale (9B). Transfer to ARC/GSM8K needs no refitting. Every operator is  $O(d)$  per token with 19 KB state (Prop. 2: optimal order).

## §2. Related Work

Linear steering. ActAdd [19] and CAA [13] add fixed diff-in-means vectors; ITI [9] shifts probe-selected heads; RepE [24] generalizes reading/control vectors; directional ablation removes one direction everywhere [1]; SAE clamping fixes a feature value [18]. All are special cases of our transfer-function family, and all are input-independent edits. AxBench [22] finds them weaker than prompting, so a new method must justify itself on locality, dose control, and serving cost, which we formalize. CAST [8] adds conditional gating, but its action remains additive; SADI [21] and KL-adaptive composition [16] adapt dose per input.

Distribution matching and optimality. MiMiC [17] derives the Gaussian- $W_2$  affine map, applied unconditionally in the full space; AcT [14] fits per-neuron 1-D transport; LinEAS [15] learns affine maps under a distributional loss. COAST [12] proves pointwise optimality of a covariance-weighted projection; in our language, transport to a point mass (their  $\Sigma$ -metric is the  $M$  of Prop. 1); A-LQR [11] proves LQR-optimality of a control input. Our Theorem 1 is a different statement: the optimal map between distributions under a locality constraint, with the conditional optimality and selectivity ceiling (Thms. 2–3) that no prior steering work states. PCHI [10] trains probe-conditioned head rescalings for the same behavior; its additive baseline corrects  $\approx 0\%$ , independently confirming Cor. 1. [23] reports the dose-response our per-sample dose exploits.

### §3. Method: Projection-Transport Steering

Let  $h \in \mathbb{R}^d$  be the residual-stream activation at a fixed decoder layer and let  $V \in \mathbb{R}^{k \times d}$  have orthonormal rows spanning a behavioral subspace (here  $k \leq 2$ : a confidence axis and a correctness axis obtained by diff-in-means on the model’s own labeled reasoning traces). Write  $q = Vh \in \mathbb{R}^k$ , let  $\mu$  be the law of  $h$  under natural generation,  $\mu_V$  the induced law of  $q$ , and  $\nu$  a target law on  $\mathbb{R}^k$ .

**Definition 1** (admissible steering).  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  is admissible for  $(V, \nu)$  if (i)  $(I - V^\top V)T(h) = (I - V^\top V)h$   $\mu$ -a.s. (locality: the orthogonal complement is untouched), and (ii)  $(V \cdot T)_\# \mu = \nu$ , i.e. the pushforward of  $\mu$  equals the target law (goal: steered projections follow the target law). The cost is  $C(T) = \mathbb{E}_\mu \|T(h) - h\|^2$ .

The cost is the right regularizer: large interventions are what empirically destroy generation (Sec. 5), and under a quadratic functional metric  $C$  bounds the KL drift of the next-token distribution (Prop. 1).

**Theorem 1** (steering is optimal transport of the projection).  $\min_T C(T) = W_2^2(\mu_V, \nu)$  over admissible  $T$ , attained by  $T^*(h) = h + (S(Vh) - Vh)^\top V$  where  $S$  is the ( $\mu_V$ -a.s. unique) optimal Monge map from  $\mu_V$  to  $\nu$ . Any optimal  $T$  satisfies  $VT(h) = S(Vh)$  a.s.; in particular the optimal intervention is a deterministic function of the projection alone; conditioning on  $h_\perp$  cannot reduce the cost.

**Proof sketch.** Locality collapses the cost to the projection;  $(Vh, VT(h))$  couples  $(\mu_V, \nu)$ , so  $C(T) \geq W_2^2$ ; Brenier’s map attains it. Full proof in Appendix B.  $\square$

For  $k = 1$ ,  $S = F_\nu^{-1} \circ F_{\mu_V}$  is the monotone quantile map, implemented as a 41-knot piecewise-linear lookup. For Gaussian marginals the map is affine (Knott–Smith [7]; Gelbrich [4]):  $S(q) = m_t + A(q - m_s)$  with  $A = \Sigma_s^{-1/2} (\Sigma_s^{1/2} \Sigma_t \Sigma_s^{1/2})^{1/2} \Sigma_s^{-1/2}$ .

Corollary 1 (additive steering is forced to be global). Additive steering  $g(p) = p + \alpha$  is the optimal map iff  $\nu = \mu_V * \delta_\alpha$  (pure shift). If the target compresses the spread ( $\sigma_t < \sigma_s$ ; what “hedge only abnormal confidence” requires), the optimal slope is  $\sigma_t/\sigma_s < 1$ : the tail must move more than the bulk. A slope-1 map matched to the tail over-transportes the bulk by the same constant; the observed CR collapse. Moreover its selectivity between two subpopulations with equal projection laws is identically zero.

Theorem 2 (optimal gate; selectivity ceiling). Let a gate fire iff a trace statistic  $s > \tau$ , with  $\text{TPR}(\tau) = P(s > \tau \mid OCW)$ ,  $\text{FPR}(\tau) = P(s > \tau \mid CR)$ . (a) For any objective increasing in TPR and decreasing in FPR, the likelihood-ratio test is optimal at every operating point (Neyman–Pearson); under equal-covariance Gaussian class conditionals it reduces to the linear discriminant  $w = \Sigma^{-1}(\mu_{OCW} - \mu_{-OCW})$ . (b) For any statistic with concavified ROC area AUC,  $\max_\tau [\text{TPR} - \text{FPR}] \leq 2 \text{AUC} - 1$ .

Theorem 3 (joint optimality of gate  $\times$  transport). Among interventions that are local to  $V$ , identity off the gate event  $\Gamma = \{s > \tau\}$ , and achieve target law  $\nu_\Gamma$  on  $\Gamma$ , the cost-minimal intervention applies Theorem 1’s map for the conditional law  $(\mu_V \mid \Gamma) \rightarrow \nu_\Gamma$  on  $\Gamma$ ; its cost is  $P(\Gamma) W_2^2(\mu_V \mid \Gamma, \nu_\Gamma)$ .

Proposition 1 (metric covariance). If cost is  $(\Delta h)^\top M (\Delta h)$  with  $M \succ 0$  (e.g. a Fisher/Gauss–Newton pullback, so cost approximates the induced next-token KL), the substitution  $y = M^{1/2} h$  reduces the problem to Theorem 1; Euclidean PTS is  $M = I$ . Proof in Appendix B.

Remark 1 (support preservation). The quantile map never leaves the target’s empirical support; an additive map violates support for every  $\alpha \neq 0$ , consistent with the observed generation collapse at  $|\alpha| = 1.5 \|h\|$ .

Proposition 2 (computational optimality). Any algorithm implementing a non-degenerate intervention local to a subspace spanned by a dense direction must read all  $d$  coordinates of  $h$ , i.e. costs  $\Omega(d)$  per token. PTS costs  $\Theta(kd + \log K)$ ; matching the lower bound for constant  $k, K$ ; versus  $\Theta(d^2)$  for a full-space affine map (MiMiC),  $\Theta(d d_{\text{sae}})$  for SAE steering, and a full extra forward pass for KL-adaptive dosing. Proof in Appendix B.

## §4. Experimental Setup

Models, datasets, and splits. The main experiments use gemma-2-2b-it [5]; the central comparison is repeated on gemma-2-9b-it at  $4.5\times$  scale (Sec. 6).<sup>1</sup> Both models reason inside `<think>...</think>` and answer `\boxed{letter}`; decoding is greedy, so evaluation “seeds” resample questions, not decoding noise. Activations are read and modified at layer 14/26 (2B) and 24/42 (9B). Benchmarks: MMLU [6] (fitting and primary evaluation), ARC-Challenge [2], and GSM8K [3] as 4-option MCQ with seeded distractors (GSM8K-MCQ, Appendix C);  $n=300$  questions each, disjoint from all fitting data. Directions, maps, and gate parameters are fit once per scale on a 400-question MMLU extraction split ( $\sim 50k$  trace tokens) and reused everywhere: cross-benchmark results are transfer, not refits. The main comparison uses five resampled MMLU subsets; the 9B replication three.

Behavioral states and confidence readouts. Every answer gets one of four labels; overconfident-wrong (OCW), nonconfident-wrong, nonconfident-right, confident-right (CR); under two independent readouts: the letter-logit readout M2 (softmax of the letter logits at the trace’s own answer position) and the yes-probability readout M4 (normalized  $P(\text{YES})$  over per-option yes/no reformulations). On identical questions they disagree sharply (ECE 0.39 vs 0.10; Appendix D): M2 is strongly overconfident, M4 much better calibrated. M4 is therefore the primary readout; agreement under M2 and the M3 self-report is a robustness requirement. Full grids: Appendix G.

Directions and the trace-level gate. On the extraction split we fit a confidence axis  $\hat{v}$  (the steering axis) and a detection axis  $\hat{u}$ , both diff-in-means, orthonormalized into  $V = (\hat{v}, \hat{u})$ . A completed trace of length  $T$  is summarized by its mean projections  $z = (\frac{1}{T} \sum_t \langle h_t, \hat{v} \rangle, \frac{1}{T} \sum_t \langle h_t, \hat{u} \rangle)$ ; an LDA head gives the gate score  $s(z) = w^\top z + b$ , firing iff  $s(z) > \tau$ . Gated conditions are two-pass: generate unsteered; if the gate is silent, keep that output; else regenerate with the intervention active during reasoning. The gate sees only the model’s own activations; no gold labels at inference.

Steering methods. All operators act through  $p_t = \langle h_t, \hat{v} \rangle$  (Sec. 3). Additive:  $h'_t = h_t + \alpha \hat{v}$ ,  $\alpha = -0.75 \|h\|$  (the dose that eliminates OCW under the calibrated readout). Ablation:  $h'_t = h_t - p_t \hat{v}$ . Clamp:

<sup>1</sup>2B weights were served from the unsloth/gemma-2-2b-it mirror; we verified all 288 tensors are bit-identical to the official google/gemma-2-2b-it release.

$p'_t = \min(p_t, c)$  at a source quantile (median = representative condition). Quantile/Gaussian/BW transport: the monotone quantile or affine Gaussian/Bures–Wasserstein map onto a target law fitted on the extraction split. Conditional variants: the trace gate composed with each action (CAST-style gate+additive, gated clamp, gated ablation, gated OT, LDA-gated subspace BW); the gate controls exposure, the action severity, especially on false positives. Published baselines: prompting, additive CAA [13], ablation [1], CAST-style [8], MiMiC [17], Linear-Act per-neuron transport [14]; all reimplemented and run on identical records, decoding, readouts, and metrics. Think-only variants disable steering before the answer tokens.

Metrics and statistical protocol. Each metric answers one question about the intervention.

Accuracy: did it hurt task performance? We report the change  $\Delta\text{acc}$  against the unsteered model.

ECE (expected calibration error): can the confidence numbers be trusted? The mean absolute gap between average stated confidence and actual accuracy over 10 confidence bins (lower is better; 0 = confidence matches reality).

State histogram: where did the answers move? The counts of the four behavioral states before and after intervention.

Surgical metrics: did it hit only its target? Pairing answers by question,  $\text{ocw\_rm} = P(\text{left OCW} \mid \text{was OCW})$  is the fraction of the harmful state removed,  $\text{cr\_keep} = P(\text{stayed CR} \mid \text{was CR})$  the fraction of the healthy state preserved, and selectivity  $\text{Sel} = \text{ocw\_rm} - (1 - \text{cr\_keep})$  combines them (+1 = perfect scalpel; 0 = indiscriminate).

Selective prediction (AURC/E-AURC): does confidence still rank right answers above wrong ones? AURC is the area under the risk–coverage curve (lower is better); monotone post-hoc recalibration leaves it invariant, so only interventions that change answers or their ranking can move it.

Single-subset results carry paired-bootstrap 95% CIs; multi-subset results are  $\text{mean} \pm \text{std}$  over independently resampled subsets.

Post-review validation protocol. We froze a simplified detector-axis action after a development-only Qwen3 component analysis, then evaluated it on held-out TruthfulQA panels for Mistral-7B, OLMo-2-7B, and Granite-8B. Each eligible architecture used the same 237-question composition (50 in-domain, 187 shifted) and was compared with a norm-matched spherical

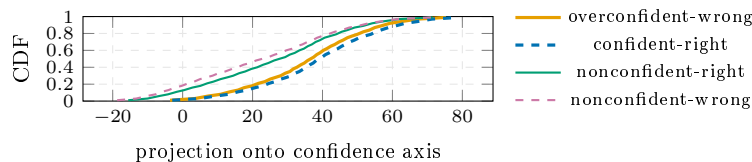


Figure 1. Token-level projection CDFs on the confidence axis by behavioral state (extraction split,  $\sim 50k$  tokens). The OCW and CR curves are nearly identical (Cohen's  $d = 0.049$ ); the confident/nonconfident pairs are well separated ( $d \approx 0.6$ ).

action using 10,000 paired BCa resamples; the three architecture comparisons used Bonferroni-adjusted two-sided intervals. OLMo-2 subsequently faced seven pre-frozen construction and specificity controls. A separate six-cell Mistral/OLMo-2/Granite  $\times$  BBQ/ETHICS observer screen kept fitting, screening, and confirmatory units disjoint; only Mistral/ETHICS advanced to a 1,000-scenario confirmatory test, with intervals adjusted for the six planned cells. Only observer-eligible cells were opened; the planned denominator remained six.

## §5. Results on MMLU

The confidence axis does not separate justified from unjustified confidence. Figure 1 tests the geometric premise of selective steering: the OCW and CR projection CDFs nearly coincide ( $d = 0.049$ ) while the confident/nonconfident pairs separate ( $d \approx 0.6$ ). The axis encodes whether the model is confident, not whether the confidence is warranted, so no transfer function of this coordinate can be surgical (Cor. 1; Thm. 2b,  $AUC \approx .51$ ). The measurements agree: the selected dose removes all 52 OCW but takes CR from 77 to 1; across five subsets additive selectivity is  $+0.003 \pm 0.006$ , accuracy drops  $7.8 \pm 1.5$  points, AURC worsens  $0.410 \rightarrow 0.534$ . Eliminating OCW is not evidence of calibration: the shift suppresses desirable and undesirable confidence alike.

Projection-dependent actions reduce collateral damage. Median clamping; one scalar touching only the abnormal tail; reaches selectivity  $+0.234$  on the representative subset at  $+5.3$  accuracy points; across five subsets accuracy is  $+0.007 \pm 0.032$ , i.e. preserved, not improved (single-split accuracy gains should be treated as noise until seed-replicated), while selectivity holds at  $+0.199 \pm 0.071$ . Quantile transport to the non-OCW target reaches  $+0.141 \pm 0.083$ , likewise without the additive loss. The

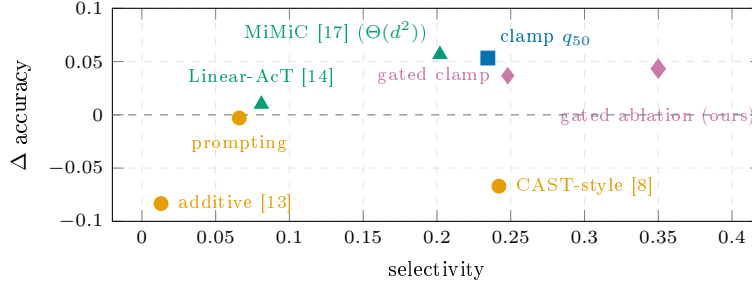


Figure 2. The selectivity–accuracy frontier, identical MMLU protocol (M4, representative subset; multi-subset means in Tab. 1). Additive-action methods (orange) pay accuracy for selectivity or achieve neither; ungated distribution matching (green) is safe but weakly selective; gate  $\times$  tail-local action (ours) dominates at  $O(d)$  per-token cost.

| method (M4, MMLU, seeds mean $\pm$ std) | $\Delta$ acc       | ECE               | AURC              | Sel                |
|-----------------------------------------|--------------------|-------------------|-------------------|--------------------|
| baseline                                | —                  | 0.110 $\pm$ 0.021 | 0.410 $\pm$ 0.014 | —                  |
| additive $-0.75$ [13]                   | $-0.078 \pm 0.015$ | 0.112 $\pm$ 0.016 | 0.534 $\pm$ 0.026 | $+0.003 \pm 0.006$ |
| CAST-style gate+add [8]                 | $-0.043 \pm 0.029$ | 0.100 $\pm$ 0.018 | 0.448 $\pm$ 0.039 | $+0.214 \pm 0.111$ |
| MiMiC full-space affine [17]            | $+0.009 \pm 0.048$ | 0.110 $\pm$ 0.017 | 0.391 $\pm$ 0.056 | $+0.194 \pm 0.084$ |
| clamp $q_{50}$ (ours)                   | $+0.007 \pm 0.032$ | 0.104 $\pm$ 0.012 | 0.404 $\pm$ 0.034 | $+0.199 \pm 0.071$ |
| quantile-OT non-OCW (ours)              | $-0.005 \pm 0.019$ | 0.102 $\pm$ 0.018 | 0.415 $\pm$ 0.032 | $+0.141 \pm 0.083$ |
| gated clamp (ours)                      | $+0.011 \pm 0.027$ | 0.109 $\pm$ 0.012 | 0.405 $\pm$ 0.028 | $+0.208 \pm 0.106$ |
| gated ablation (ours)                   | $+0.009 \pm 0.036$ | 0.093 $\pm$ 0.021 | 0.387 $\pm$ 0.045 | $+0.266 \pm 0.112$ |
| LDA-gated ablation (ours)               | $+0.013$           | 0.068             | 0.375             | $+0.261$           |

Table 1. Multi-seed head-to-head (five evaluation-subset seeds,  $n=300$  each; greedy decoding is deterministic, so seeds resample questions; a stricter robustness axis than sampling seeds). Only Linear-AcT reports seed variance in this literature.

lesson is shape, not strength: abnormal high-confidence activations must move more than ordinary ones, which ablation, clamp, and transport achieve by conditioning on the input projection.

Main result. Across five MMLU subsets our gated ablation attains selectivity  $+0.266 \pm 0.112$  (additive  $+0.003 \pm 0.006$ , prompting  $+0.066$ ) at no detectable accuracy cost ( $+0.009 \pm 0.036$ ), with the best ECE ( $.093 \pm .021$ ) and AURC ( $.387 \pm .045$ ), beating faithful reimplementations of CAST-style, MiMiC, and Linear-AcT on every column (Table 1, Fig. 2).

CAST-style gate+additive reaches  $+0.214 \pm 0.111$  but pays  $4.3 \pm 2.9$  accuracy points on every subset and pushes AURC above baseline (.448 vs .410): the gate improves targeting, but the additive action inside still over-transportes (Cor. 1). MiMiC preserves accuracy but is weaker ( $+0.194 \pm 0.084$ ) and rewrites the full residual stream at  $\Theta(d^2)$  per token ( $2304\times$  our cost). Prompting reaches  $+0.066$  and Linear-AcT  $+0.07-0.08$  (representative subset). Per-seed, gated ablation beats CAST-style 3/5 on selectivity alone, 5/5 counting accuracy, and MiMiC 5/5 (mean paired gap  $+0.072$ ). The claim is not per-metric maxima in every run but the strongest joint trade-off across removal, retention, accuracy, and selective prediction (Fig. 2).

Gate and action shape are complementary. Figure 3 (Appendix E) shows why gate and local action compose, by the measured movement of real regenerated traces. Additive steering moves all 52 OCW far past the calibrated zone and drags all 77 CR along. Gated ablation protects twice: 29 CR below threshold stay bitwise identical (74/300 questions overall), and of the 48 CR it wrongly flags, 32 keep their state because the tail-local action barely moves typical activations. CAST-style has the first protection but not the second; ungated clamp/transport the second but not the first.

Detection is the main remaining bottleneck. The LDA gate separates OCW from CR at trace-level AUROC  $\approx 0.77$  (catches 45/52, misses 7, false-fires on 48/77 CR). The diagnostic value  $2\text{AUROC} - 1 \approx 0.538$  is close to twice the achieved selectivity; Thm. 2b applies to the concavified ROC area, so raw AUROC is not an exact behavioral ceiling. Detection, rather than the transport map, is nevertheless the binding empirical constraint; Fig. 4 (bottom) plots the gap against the label oracle.

Robustness across instruments; think-only steering. Absolute counts vary across readouts, but the conclusions persist: gated ablation reaches selectivity  $+0.35$  (M4),  $+0.15$  (M2),  $+0.17$  (M3 self-report: 37% of OCW removed at 0.80 retention). Think-only variants keep every effect’s direction at near-zero accuracy cost: the intervention acts through the reasoning trajectory, not the answer logits.

## §6. Scale, Transfer, and Serving Cost

Replication at 9B. The 2B story replicates on gemma-2-9b-it (Table 2): additive steering has identically zero selectivity and CR-retention on every subset while losing  $18.4 \pm 3.3$  points; CAST-style buys  $+0.283$  selectivity for  $11.4 \pm 3.2$  points and 0.45 retention; the local operators alone gain

| method (M4, 9B, 3 seeds) | $\Delta_{\text{acc}}$ | cr_keep           | AURC              | Sel                |
|--------------------------|-----------------------|-------------------|-------------------|--------------------|
| baseline                 | —                     | $0.144 \pm 0.021$ | $0.197 \pm 0.025$ | —                  |
| additive $-0.75$         | $-0.184 \pm 0.033$    | $0.000 \pm 0.000$ | $0.450 \pm 0.023$ | $+0.000 \pm 0.000$ |
| CAST-style gate+add      | $-0.114 \pm 0.032$    | $0.453 \pm 0.038$ | $0.285 \pm 0.016$ | $+0.283 \pm 0.093$ |
| clamp $q_{50}$ (ours)    | $+0.023 \pm 0.012$    | $0.887 \pm 0.015$ | $0.184 \pm 0.025$ | $+0.202 \pm 0.020$ |
| gated ablation (ours)    | $+0.011 \pm 0.008$    | $0.907 \pm 0.014$ | $0.186 \pm 0.023$ | $+0.203 \pm 0.034$ |

Table 2. gemma-2-9b-it, MMLU, layer 24,  $n=300 \times 3$  seeds:  
the head-to-head replicates at  $4.5\times$  scale.

| condition (M4)       | MMLU                  |       |            | ARC-Challenge         |       |            | GSM8K (MCQ)           |       |          |
|----------------------|-----------------------|-------|------------|-----------------------|-------|------------|-----------------------|-------|----------|
|                      | $\Delta_{\text{acc}}$ | ECE   | Sel        | $\Delta_{\text{acc}}$ | ECE   | Sel        | $\Delta_{\text{acc}}$ | ECE   | Sel      |
| baseline             | —                     | 0.123 | —          | —                     | 0.092 | —          | —                     | 0.280 | —        |
| additive $-0.75$     | $-0.083^*$            | 0.099 | $+0.013$   | $-0.123^*$            | 0.265 | $+0.000$   | $-0.257^*$            | 0.069 | $+0.000$ |
| clamp $q_{50}$       | $+0.053^*$            | 0.116 | $+0.234^*$ | $+0.037$              | 0.116 | $+0.197^*$ | $+0.007$              | 0.286 | $-0.083$ |
| quantile-OT non-OCW  | $+0.013$              | 0.075 | $+0.158^*$ | $-0.027$              | 0.073 | $+0.095$   | $+0.010$              | 0.278 | $+0.083$ |
| gated clamp $q_{50}$ | $+0.037$              | 0.105 | $+0.248^*$ | $+0.027$              | 0.124 | $+0.137$   | $+0.020$              | 0.295 | $+0.167$ |

Table 3. Zero-shot transfer of MMLU-fitted steering ( $n=300$  each;  $*$  = 95% bootstrap CI excludes 0).

accuracy and improve selective prediction (AURC 0.184–0.186 vs 0.197), at selectivity  $+0.20$  and retention 0.89–0.91. Scale makes the global shift costlier, not more selective.

Cross-domain transfer. Table 3 applies the MMLU-fitted directions, maps, and thresholds to ARC-Challenge and GSM8K-MCQ with no re-fitting; Fig. 4 (top) shows the two geometries. ARC-Challenge carries a real overconfident tail and the operators transfer: clamp keeps  $+0.197$  selectivity (CI excludes 0) at  $\Delta_{\text{acc}} +0.037$ ; gated ablation transfers at  $+0.222$  [0.05, .39],  $\Delta_{\text{acc}} -0.003$ , AURC .180. GSM8K-MCQ has only 6/300 baseline OCW: the tail targeted on MMLU is absent, the input-dependent maps correctly degenerate to the identity ( $\Delta_{\text{acc}} \approx 0$ ; selectivity CIs are wide), while the fixed shift subtracts 25.7 points there and 12.3 on ARC with selectivity pinned at zero. Threshold transfer is the open limitation: an absolute MMLU threshold can fire on none or all of a target domain; label-free re-anchoring of  $\tau$  to a domain gate-score quantile fixes it (Appendix G).

Post-review detector-axis validation. Table 4 reports the positive pre-declared held-out check. On OLMo-2/TruthfulQA, the detector-axis action improves both primary metrics over a norm-matched spherical action and passes all seven pre-frozen construction controls. The effect is not explained by intervention magnitude alone, and fitting the representation at

| model / panel       | comparison          | $\Delta$ calibrated MC2 [adj. 95% CI] | $\Delta$ MC1 [adj. 95% CI] | decision |
|---------------------|---------------------|---------------------------------------|----------------------------|----------|
| OLMo-2 / TruthfulQA | matched spherical   | + .137 [+ .105, + .168]               | + .105 [+ .025, + .177]    | pass     |
| OLMo-2 / TruthfulQA | no intervention     | + .299 [+ .251, + .345]               | + .287 [+ .177, + .384]    | pass     |
| OLMo-2 / TruthfulQA | strongest isotropic | + .220 [+ .180, + .258]               | + .194 [+ .097, + .287]    | pass     |
| OLMo-2 / TruthfulQA | unprompted fit      | + .295 [+ .247, + .343]               | + .287 [+ .177, + .384]    | pass     |

Table 4. Held-out OLMo-2 validation on 237 TruthfulQA questions. Intervals use 10,000 paired BCa resamples and the predeclared Bonferroni family. The full seven-control family passes; the table shows the matched action and representative construction controls. Qwen3 was development-only and is excluded from confirmatory claims.

prompted answer positions is necessary relative to the unprompted ablation. This confirms that the detector-axis construction is not specific to Gemma.

Qualitative effects. Gated ablation does not induce generic hedging (hedge markers drop  $0.57 \rightarrow 0.27$ ; 13 OCW become correct); trace analysis and samples in Appendix F.

Serving cost. Per token and per behavior, the action uses one  $d$ -dim dot product, a 41-knot lookup, one FMA ( $\approx 4d$  FLOPs  $\approx 10^{-4}$  of a layer; Prop. 2: the optimal order); the gate adds a dot product and an  $O(1)$  running mean; state  $\approx 19$  KB. The action needs no extra forward or KV-cache invalidation; the tested post-hoc trace gate is two-pass and regenerates flagged traces. Strength interpolates as  $g_\lambda = (1 - \lambda)\text{id} + \lambda g$ ; matches vLLM residual-stream hooks [20].

## §7. Limitations and Assumptions

The optimal-transport results concern minimal activation displacement under a specified subspace, target projection law, and locality constraint; they do not establish that the chosen projection is sufficient for downstream behavior. The Monge statement also requires the usual existence and regularity conditions for a deterministic optimal map. The gate theorem bounds statistical separability of its score, while behavioral utility additionally depends on the action and model response. Its area bound uses the concavified ROC, not raw empirical AUROC.

Empirically, the five main evaluation seeds are independently resampled question subsets under deterministic decoding, not independent training runs. The original positive evidence covers overconfidence on two Gemma scales and zero-shot benchmark transfer; the held-out extension adds one confirmed OLMo-2/TruthfulQA result. Other architecture/behavior cells

did not satisfy all predeclared gates, and the advancing ETHICS cell exposed label-stratum heterogeneity (Appendix A). These boundaries preclude a universal detector-axis claim. Finally, the reported  $O(d)$  bound is an operator-cost claim; end-to-end latency for the evaluated post-hoc gate includes a second generation for flagged traces.

## §8. Conclusion

We turned an empirical steering failure into theorems and a method: the additive edit is the wrong shape for tail-compression targets; optimal transport along a behavioral subspace is the provably minimal intervention; selectivity is constrained by detection as well as action response; and the NP-gate  $\times$  tail-local action gives the strongest joint trade-off in our Gemma/MMLU comparison across five subsets and two model scales, at no detectable accuracy cost and minimal per-token action compute. Held-out validation on OLMo-2/TruthfulQA supports the detector-axis construction beyond Gemma. Future work: better detection axes (at 9B AUROC rises to 0.83), Fisher-metric transport, further behaviors.

## §9. Contributions

Mark Kashirskiy: problem formulation of surgical steering; the PTS framework, Theorems 1–4, Prop. 2 and proofs; all steering operators, baseline reimplementations, and experiments (2B and 9B, five seeds, four benchmarks); visualizations; serving analysis; paper draft. Daniil Sergeev: the confidence measurement suite (M1–M5) and its  $n=1000 \times 3$  study; feature-extraction pipeline (CAA/probe/SAE); the original steering grid that exposed the global-hedging failure; the 9B measurement replication; compute infrastructure. Tatevik Ter-Hovhannisyanyan: the proposal of conditional (gated) steering; behavioral labeling methodology and evaluation protocols (jointly with D.S.: the LLM-as-judge and labeled-dataset design for the measurement suite and the next behavioral domains); initial text review of the paper; project direction and coordination. Irina Piontkovskaya (mentor): project direction, research supervision, methodology review; we thank her for guidance throughout the project.

## References

1. Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda, Refusal in language models is mediated by a single direction, Advances in Neural Information Processing Systems 37, 2024.

2. Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord, Think you have solved question answering? Try ARC, the AI2 reasoning challenge, arXiv:1803.05457, 2018.
3. Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, et al., Training verifiers to solve math word problems, arXiv:2110.14168, 2021.
4. Matthias Gelbrich, On a formula for the  $L^2$  Wasserstein metric between measures on Euclidean and Hilbert spaces, *Mathematische Nachrichten* 147 (1990), no. 1, 185–203.
5. Gemma Team, Gemma 2: Improving open language models at a practical size, arXiv:2408.00118, 2024.
6. Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt, Measuring massive multitask language understanding, *International Conference on Learning Representations*, 2021.
7. Martin Knott and Cyril S. Smith, On the optimal mapping of distributions, *Journal of Optimization Theory and Applications* 43 (1984), 39–49.
8. Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehling, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar, Programming refusal with conditional activation steering, *International Conference on Learning Representations*, 2025.
9. Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg, Inference-time intervention: Eliciting truthful answers from a language model, *Advances in Neural Information Processing Systems* 36, 2023.
10. Ke Li, Chongzhe Zhang, Zifan Zeng, Feng Liu, Qunli Zhang, and Zheng Hu, Calibrating overconfidence without sacrificing confidence: Probe-conditioned head intervention for LLMs, arXiv:2606.09876, 2026.
11. Julian Skifstad, Xinyue Annie Yang, and Glen Chou, Local linearity of LLMs enables activation steering via model-based linear optimal control, arXiv:2604.19018, 2026.
12. Tam Nguyen, Tu Anh Nguyen, Sina Alemohammad, and Richard G. Baraniuk, Minimizing collateral damage in activation steering, arXiv:2605.01167, 2026.
13. Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner, Steering Llama 2 via contrastive activation addition, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
14. Pau Rodriguez, Arno Blaas, Michal Klein, Luca Zappella, Nicholas Apostoloff, Marco Cuturi, and Xavier Suau, Controlling language and diffusion models by transporting activations, *International Conference on Learning Representations*, 2025.
15. Pau Rodriguez, Michal Klein, Eleonora Gualdoni, Arno Blaas, Luca Zappella, Marco Cuturi, and Xavier Suau, LinEAS: End-to-end learning of activation steering with a distributional loss, arXiv:2503.10679, 2025.
16. Daniel Scalena, Gabriele Sarti, and Malvina Nissim, Multi-property steering of language models with dynamic activation composition, *Proceedings of BlackboxNLP*, 2024.
17. Shashwat Singh, Shauli Ravfogel, Jonathan Herzig, Roei Aharoni, Ryan Cotterell, and Ponnurangam Kumaraguru, Representation surgery: Theory and practice of

- affine steering, Proceedings of the 41st International Conference on Machine Learning, 2024.
18. Adly Templeton, Tom Conerly, Jonathan Marcus, et al., Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet, Transformer Circuits Thread, 2024.
  19. Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid, Steering language models with activation engineering, arXiv:2308.10248, 2023.
  20. UK AI Safety Institute, vllm-lens: activation extraction and steering for vLLM, <https://github.com/UKGovernmentBEIS/vllm-lens>, 2026.
  21. Weixuan Wang, Jingyuan Yang, and Wei Peng, Semantics-adaptive activation intervention for LLMs via dynamic steering vectors, International Conference on Learning Representations, 2025.
  22. Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts, AxBench: Steering LLMs? Even simple baselines outperform sparse autoencoders, Proceedings of the 42nd International Conference on Machine Learning, 2025.
  23. Tianyi Zhao, Yinhan He, Wendy Zheng, Yujie Zhang, and Chen Chen, Wired for overconfidence: A mechanistic perspective on inflated verbalized confidence in LLMs, Conference on Language Modeling (COLM), 2026.
  24. Andy Zou, Long Phan, Sarah Chen, James Campbell, et al., Representation engineering: A top-down approach to AI transparency, arXiv:2310.01405, 2023.

## Appendix §A. Post-review stress-test boundary

The confirmatory program preserved every planned cell in the denominator. Mistral/TruthfulQA improved calibrated MC2 by  $+0.153$  with adjusted 95% CI  $[+0.108, +0.197]$ , but its MC1 interval  $[-0.059, +0.122]$  crossed the predeclared joint gate; Granite/TruthfulQA stopped at observer eligibility. Five of six BBQ/ETHICS observer cells did not advance. Mistral/ETHICS did advance and improved pooled calibrated MC2 by  $+0.107$   $[+0.090, +0.124]$ , but its label-1 MC1 changed by  $-0.093$   $[-0.124, -0.071]$ . Thus the OLMo-2 result is a confirmed architecture transfer, while model-wide and behavior-wide generalization remain unsupported. All intervals use 10,000 paired BCa resamples with the predeclared Bonferroni family.

## Appendix §B. Full proofs

Theorem 1. By locality,  $T(h) - h \in \text{span}(V)$   $\mu$ -a.s., hence  $\|T(h) - h\|^2 = \|VT(h) - Vh\|^2$  and  $C(T) = \mathbb{E}\|VT(h) - Vh\|^2$ . Let  $\gamma$  be the joint law of  $(Vh, VT(h))$ ; by the goal constraint  $\gamma$  is a coupling of  $(\mu_V, \nu)$  and  $C(T) = \int \|y - x\|^2 d\gamma \geq W_2^2(\mu_V, \nu)$ . Since activations are continuous,  $\mu_V$  is atomless with finite second moment, so by Brenier’s theorem the optimal

coupling is unique and induced by a map  $S = \nabla\varphi$ ,  $\varphi$  convex;  $T^*(h) = h + (S(Vh) - Vh)^\top V$  is admissible and attains the bound. For any optimal  $T$ , the induced coupling must equal the unique optimal one, i.e.  $VT(h) = S(Vh)$  a.s. Since  $S$  depends only on  $Vh$ , the optimal intervention is  $\sigma(Vh)$ -measurable: access to  $h_\perp$  is useless.  $\square$

Corollary 1. If  $\nu = \mu_V * \delta_\alpha$  the monotone map is  $x \mapsto x + \alpha$ . Conversely if the optimal map is  $x + \alpha$  then  $\nu$  is the shift of  $\mu_V$ . For Gaussians with  $\sigma_t \neq \sigma_s$ , after the best shift the remaining distance is  $W_2^2(\mu_V * \delta_\alpha, \nu) = (\sigma_s - \sigma_t)^2 > 0$ . Selectivity: a map  $g$  acts identically on two subpopulations whose laws of  $p$  coincide; the paired state changes it induces then differ only through the (equal) distributions, giving equal marginal change rates, i.e. zero selectivity in expectation.  $\square$

Theorem 2. (a) is the Neyman–Pearson lemma applied to the class-conditional laws of  $s$ ; monotone-likelihood-ratio families reduce the LR test to thresholding, and for  $\mathcal{N}(\mu_B, \Sigma)$  vs  $\mathcal{N}(\mu_G, \Sigma)$  the log-LR is affine in  $s$  with direction  $\Sigma^{-1}(\mu_B - \mu_G)$  (LDA). (b) The concavified ROC contains the segment path  $(0, 0) \rightarrow (\text{FPR}^*, \text{TPR}^*) \rightarrow (1, 1)$  through the Youden-optimal point; its area is  $\frac{1}{2}(1 + \text{TPR}^* - \text{FPR}^*) = \frac{1+J}{2} \leq \text{AUC}$ .  $\square$

Theorem 3. Decompose  $C(T) = P(\Gamma)\mathbb{E}[\|\Delta\|^2 \mid \Gamma] + P(\Gamma^c)\mathbb{E}[\|\Delta\|^2 \mid \Gamma^c]$ . Identity off  $\Gamma$  zeroes the second term and is forced by the constraint; the first term with the conditional goal constraint is exactly the Theorem 1 problem for  $\mu(\cdot \mid \Gamma)$ . The constraints and objective separate over the partition  $\{\Gamma, \Gamma^c\}$ , so the piecewise optimum is global.  $\square$

Proposition 1. With  $y = M^{1/2}h$ ,  $\|\Delta h\|_M^2 = \|\Delta y\|^2$ ; locality w.r.t.  $V$  becomes locality w.r.t.  $M^{1/2}V$  (a linear subspace), and pushforward constraints transform covariantly under the isomorphism. Apply Theorem 1 in  $y$ -space and map back.  $\square$

Proposition 2. Lower bound: let  $\hat{v}_j$  be dense ( $(\hat{v}_j)_i \neq 0$  for all  $i$ ) and  $g$  non-constant in  $p_j$ . If an algorithm leaves coordinate  $i$  unread, an adversary perturbs  $h_i$ ; this changes  $p_j = \langle h, \hat{v}_j \rangle$  and hence, on a set of positive measure, the correct output  $T(h)$ , while the algorithm’s output is unchanged; contradiction. Learned diff-in-means directions are dense almost surely. Upper bound: PTS computes  $p = Vh$  ( $kd$  multiply-adds), a monotone  $K$ -knot lookup ( $O(\log K)$  by binary search), and  $h + (g(p) - p)^\top V$  ( $kd$  more):  $\Theta(kd + \log K)$  total. Memory  $\Theta(kd + kK)$ : reproducing a target marginal at quantile resolution  $1/K$  requires  $\Omega(K)$  parameters, and specifying the subspace requires  $kd$ .  $\square$

### Appendix §C. GSM8K MCQ construction

Each GSM8K test item with integer gold  $g$  receives three deterministic distractors drawn (seeded by item index) from  $\{g \pm 1, g \pm 2, g \pm 10, 2g, \lfloor g/2 \rfloor, \text{round}(1.5g), \text{round}(0.8g), g \pm \text{U}\{3..9\}\}$ , deduplicated, shuffled with the gold into four options. This converts free-form arithmetic into selection while keeping the reasoning burden; we report transfer numbers on it and do not claim GSM8K free-form results.

### Appendix §D. The behavior across domains and scales (measurement study)

Overconfidence is model  $\times$  instrument (2B). Five confidence readouts on identical MMLU questions ( $n=1000 \times 3$  seeds) disagree categorically: ECE ranges from .391 [.374, .409] (letter-logit M2, the literature default) through .249/.177 (self-report/answer-distribution) to .104 [.086, .120] (per-option yes/no M4); disjoint CIs; the instruments even disagree on the answer (all five agree on 28.9% of questions). Hence our protocol requires every steering claim to hold under multiple readouts. Domains: on MMLU the model is overconfident (52/300 OCW), on GSM8K-MC and GPQA underconfident (6 and 27 OCW); the failure mode is domain-dependent, and the input-dependent operators inherit exactly this dependence (Table 3).

Scale replication (9B). The full pipeline was replicated on gemma-2-9b-it (42 layers,  $d=3584$ , layer 24) over the same domains. Method-dependence survives scale (MMLU ECE .126–.266, GPQA .177–.499, same instrument ranking); scale polarizes rather than cures (GSM8K nearly solved and calibrated, gap  $-0.015$ ; GPQA confidence grows faster than knowledge, M2 conf .86 at acc .36); the confidence cone replicates (cosines 0.63–0.83); confidence becomes more linearly readable (detector AUROC to 0.83 vs 0.62–0.67 at 2B), so the Thm. 2b ceiling rises with scale; and steering remains one-directional, as Remark 1 predicts.

### Appendix §E. What each method does: measured movement of real traces

Where the benchmarks live; ours vs the label oracle. Figure 4 (left) plots each transfer benchmark’s baseline traces in the same plane: MMLU and ARC carry a real overconfident-wrong mass in the gated (upper-right) region, while GSM8K and GPQA have almost none; the geometric reason the

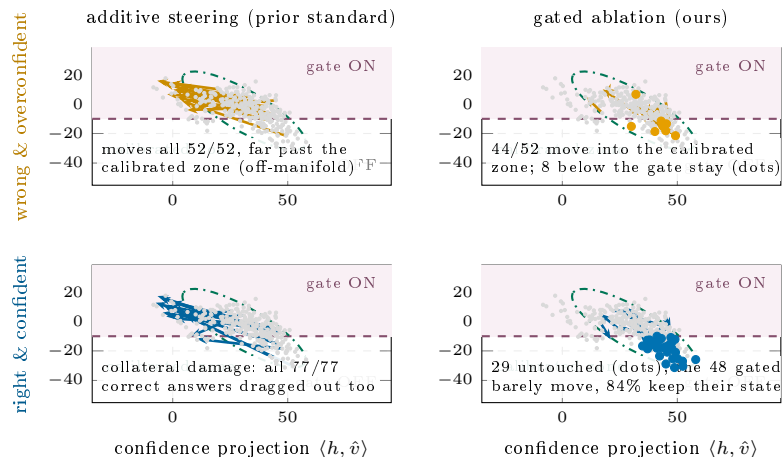


Figure 3. Each dot is one reasoning trace before steering; an arrow is its measured movement after steering (real rollouts); a bold dot without an arrow was left untouched; pink band = gate ON, green ellipse = calibrated zone ( $2\sigma$ ). Additive steering (left) overshoots all 52 OCW far past the calibrated zone and drags all 77 CR along; gated ablation (right) moves OCW into the zone and leaves 29 CR bitwise untouched.

transfer Table 3 shows activity on the former and near-identity on the latter. Figure 4 (right) compares our intervention to the label oracle; the theoretically optimal conditional steering that moves exactly the overconfident-wrong points onto the calibrated centroid: our gate catches 45/52 of the oracle’s targets (recall 0.87), misses 7, and false-fires on 48/77 confident-right points (which the tail-local action then mostly spares: 32/48 keep their state). The remaining gap to the oracle is exactly the detection gap of Thm. 2b.

#### Appendix §F. Sample traces before and after gated ablation

On the 226/300 gated questions, gated ablation does not induce hedging: hedge markers drop  $0.57 \rightarrow 0.27$  per trace, length  $95 \rightarrow 78$  words; the answer changes on 83/226 and 13 OCW become correct. Typical outcomes: an associative wrong argument replaced by an explicit derivation; better-calibrated reasoning with the same letter; a flagged CR rephrased

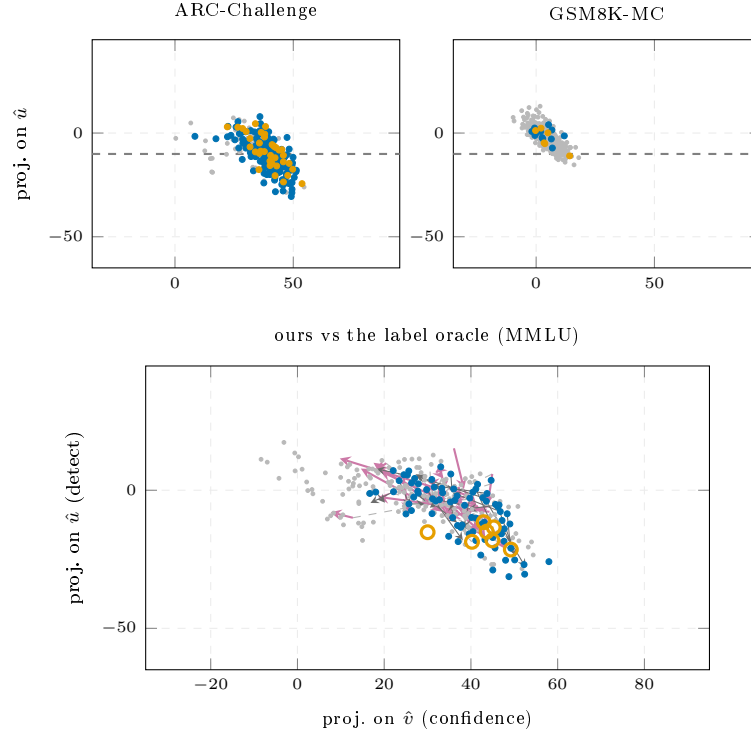


Figure 4. Top: transfer benchmarks in the behavioral plane (gate threshold dashed): ARC has an overconfident-wrong tail above the gate, GSM8K almost none; transfer activity follows the geometry. Bottom: gray dashed arrows = the label oracle (move every overconfident-wrong point to the calibrated centroid); pink = our measured moves on the same points (45/52 caught); hollow orange circles = the 7 misses; dark-gray thin arrows = false fires on confident-right (mostly state-preserving). The oracle–ours difference is the detection gap that bounds selectivity (Thm. 2b).

but preserved (strong additive/CAST doses instead yield repetition loops). Full traces: Appendix F.

Across the 226/300 gated questions, the intervention does not inject hedging vocabulary — hedge-marker frequency drops ( $0.57 \rightarrow 0.27$  per

trace) — and traces get shorter (95  $\rightarrow$  78 words). What changes is the mode of reasoning: assertive shortcut arguments are replaced by definitional, evidence-first ones, and the final answer changes on 83/226 gated questions (13 overconfident-wrong become correct under M4; 32/48 gated confident-right survive intact). Ablating the confidence projection during the think phase appears to remove a “commit early” pressure rather than adding doubt. Three representative pairs (traces truncated):

Overconfident-wrong  $\rightarrow$  corrected (mmlu-4592; gold C , M4 state overconfident wrong  $\rightarrow$  nonconfident wrong). The baseline trace jumps from a vague association to an answer; the steered trace states the definition and derives the (gold) answer.

Question: If the price of firm A's cell phone service rises by 5 percent and the quantity demanded for firm B's cell phone service increases by 10 percent, we can say that Options: A. demand for firm B is price elastic. B. supply

before: <think>The question asks about the relationship between the price of firm A's service and the quantity demanded for firm B's service. We know that the price of firm A's service increased and the quantity demanded for firm B's service increased. This suggests that the demand for firm B's service is likely to be price elastic. </think> \boxed{A}  $\rightarrow$  A

after: <think>The cross-price elasticity of demand measures the percentage change in the quantity demanded of one good in response to a percentage change in the price of another good. Since the price of firm A's service increased and the quantity demanded for firm B increased, the cross-price elasticity is positive. We can conclude that the two goods are substitutes. </think> \boxed{C}  $\rightarrow$  C

Overconfident-wrong  $\rightarrow$  calibrated-right (mmlu-12923; gold C , M4 state overconfident wrong  $\rightarrow$  nonconfident right). The baseline burns its budget re-asserting; the steered trace weighs all four options tersely and lands on the same letter with calibrated confidence.

Question: The best conclusion that can be drawn about the four-day (compressed) workweek is that it has: Options: A. long-lasting positive effects on attitudes and productivity. B. long-lasting negative effects on attitudes and p

before: <think>The question asks for the best conclusion about the four-day workweek. We need to consider the potential effects on attitudes and productivity. There's no mention of specific evidence or studies, so we need to consider the general impact of a compressed workweek. </think> <think>Option A is too strong. We don't have enough information to say it has long-lasting positive effects on attitudes and productivity. </think> <think>Option B is too strong. We don't have enough information to say it has long-las  $\rightarrow$  D

after: <think>The question asks for the best conclusion about the four-day workweek. We need to consider the potential effects on attitudes and productivity. Option A is too strong, as we don't have enough information to guarantee long-lasting positive effects. Option B is too strong, as we don't have enough information to guarantee long-lasting negative effects.

Option C is a possibility, but we don't know if it's the best conclusion.  
 Option D is a possibility, but we don't know if it's the best conclusion.  
 $\rightarrow D$

Confident-right, gated but spared (mmlu-5125; gold B, M4 state confident right  $\rightarrow$  confident right). A false-flagged correct answer: the steered reasoning changes wording, the answer improves/stays correct, and the confident-right state survives.

Question: Which one of the following best states Ivan Pavlov's accidental discovery? Options: A. While experimenting with rats, Pavlov discovered that if he repeatedly placed pellets of food in one side of a "T-maze," the rats would

before:  $\langle \text{think} \rangle$  The question asks for Pavlov's accidental discovery. Let's look at the options and see which one describes an accidental discovery. Option B describes a learned response to a stimulus, which is not an accidental discovery. Option C describes a cat learning to press a lever, which is a learned behavior. Option D describes a pigeon learning to associate a key with food. Option A describes a rat learning to run to a specific side of a maze. This is a classic example of classical conditioning. Therefore,  $\rightarrow A$

after:  $\langle \text{think} \rangle$  The question asks for the accidental discovery of Pavlov. Option B is the best choice because it describes Pavlov's accidental discovery of the dog's salivation response to the sight and sound of the attendant who brought them food.  $\rightarrow B$

One question, four methods. The most instructive comparison holds the question fixed (mmlu-4592, gold C; baseline answers A, overconfident-wrong). Our gated ablation produces the clean definitional trace shown above and lands on C. The additive-action methods do something qualitatively different — their traces degenerate into verbatim repetition loops, the signature of support violation (Remark 6): the constant shift pushes activations off the data manifold and generation cycles. Trace length balloons from 351 to  $\sim 1500$  characters of looped text; this is not deliberation but decoding pathology, invisible in aggregate confidence numbers yet obvious in the rollouts. MiMiC (distribution-matched, unconditional) stays fluent but keeps the wrong answer.

additive  $-0.75$  [13] ( $\rightarrow A$ ):

$\langle \text{think} \rangle$  The cross-price elasticity of the cell phone service is equal to the price of the cell phone service of the firm A and the quantity of the cell phone service of the firm B. The cross-price elasticity of the cell phone service is equal to the price of the cell phone service of the firm A and the quantity of the cell phone service of

CAST-style gate+add [8] ( $\rightarrow A$ ):

$\langle \text{think} \rangle$  The cross-price elasticity of demand for firm B is  $(10/5) = 2$ . The cross-price elasticity of demand for firm A is  $(10/5) = 2$ . The cross-price elasticity of demand for firm A is  $(10/5) = 2$ . The cross-price elasticity of demand for firm B is  $(10/5) = 2$ . The cross-price elasticity of demand for firm A is  $(10/5) = 2$ . The cross-price el

| condition (M4 readout)      | $\Delta$ acc | 95% CI         | ECE  | OCW 52→ | cr_keep | Sel [95% CI]             |
|-----------------------------|--------------|----------------|------|---------|---------|--------------------------|
| baseline                    | —            | —              | .123 | 52      | —       | —                        |
| additive −0.75 [13]         | −.083        | [−.15, −.02]   | .099 | 0       | .013    | + .013 [.00, .04]        |
| ablation [1]                | +.040        | [−.02, +.10]   | .083 | 36      | .623    | + .220 [.05, .39]        |
| clamp $q_{50}$ (ours)       | <b>+.053</b> | [+.003, +.103] | .116 | 40      | .792    | + .234 [.08, .41]        |
| quantile-OT non-OCW (ours)  | +.013        | [−.02, +.05]   | .075 | 44      | .831    | + .158 [.00, .32]        |
| gated clamp $q_{50}$ (ours) | +.037        | [−.01, +.09]   | .105 | 45      | .844    | <b>+ .248</b> [.09, .40] |
| gated clamp $q_{70}$ (ours) | +.003        | [−.03, +.04]   | .060 | 40      | .883    | + .210 [.06, .36]        |

Table 5. MMLU,  $n=300$ , layer 14, M4 readout (the calibrated instrument). The additive baseline removes all OCW by destroying CR (selectivity  $\approx 0$ , as Cor. 1 forces). Input-dependent transfer functions achieve positive selectivity at an accuracy gain. Same-sign effects hold under the M2 readout (Appendix G).

MiMiC [17] ( $\rightarrow$  A):

<think>The question asks about the relationship between the price of firm A's service and the quantity demanded for firm B's service. We know that the price of firm A's service increased and the quantity demanded for firm B's service increased. This suggests that the demand for firm B's service is

gated ablation (ours) ( $\rightarrow$  C):

<think>The cross-price elasticity of demand measures the percentage change in the quantity demanded of one good in response to a percentage change in the price of another good. Since the price of firm A's service increased and the quantity demanded for firm B increased, the cross-price elasticity is positive. We can conclude that the tw

## Appendix §G. Full result tables

Full condition grids under both readouts. Columns: condition, readout,  $\Delta$ accuracy with paired-bootstrap 95% CI, ECE, count of OCW after steering, CR-retention, selectivity with 95% CI. subbw = ungated subspace Bures–Wasserstein transport; npbw = LDA-gated subspace transport (self\_qXX = gate threshold re-anchored, label-free, to the target domain's own score quantile); -think = steering active only during the <think> phase.

| condition                   | r/o | $\Delta\text{acc}$ [Cl] | ECE   | OCW | cr_keep | Sel [Cl]              |
|-----------------------------|-----|-------------------------|-------|-----|---------|-----------------------|
| baseline                    | m4  | —                       | 0.123 | 52  | —       | —                     |
| act_105                     | m4  | +0.010 [−0.023, +0.047] | 0.109 | 52  | 0.86    | +0.069 [−0.06, +0.20] |
| act_110                     | m4  | +0.010 [−0.027, +0.047] | 0.113 | 46  | 0.83    | +0.081 [−0.06, +0.24] |
| cast_gate-cr_q50_add-0.75   | m4  | −0.067 [−0.123, −0.007] | 0.080 | 7   | 0.38    | +0.242 [+0.10, +0.38] |
| ablation                    | m4  | +0.040 [−0.020, +0.097] | 0.083 | 36  | 0.62    | +0.220 [+0.05, +0.39] |
| additive −0.75              | m4  | −0.083 [−0.150, −0.017] | 0.099 | 0   | 0.01    | +0.013 [+0.00, +0.04] |
| clamp_q30                   | m4  | +0.057 [+0.003, +0.113] | 0.135 | 41  | 0.64    | +0.117 [−0.05, +0.29] |
| clamp_q50                   | m4  | +0.053 [+0.003, +0.103] | 0.116 | 40  | 0.79    | +0.234 [+0.08, +0.41] |
| clamp_q70                   | m4  | +0.017 [−0.033, +0.067] | 0.091 | 46  | 0.74    | +0.106 [−0.06, +0.28] |
| gated clamp_q50             | m4  | +0.037 [−0.010, +0.087] | 0.105 | 45  | 0.84    | +0.248 [+0.09, +0.40] |
| gated OT_q50                | m4  | +0.030 [−0.007, +0.067] | 0.115 | 50  | 0.91    | +0.121 [−0.00, +0.25] |
| gated clamp_q70             | m4  | +0.003 [−0.033, +0.043] | 0.060 | 40  | 0.88    | +0.210 [+0.06, +0.36] |
| gated OT_q70                | m4  | −0.013 [−0.043, +0.020] | 0.096 | 49  | 0.94    | +0.089 [−0.02, +0.21] |
| Gaussian-OT cal.            | m4  | +0.007 [−0.030, +0.047] | 0.106 | 50  | 0.81    | +0.036 [−0.11, +0.18] |
| quantile-OT cal.            | m4  | +0.003 [−0.033, +0.040] | 0.082 | 50  | 0.88    | +0.114 [−0.02, +0.26] |
| quantile-OT non-OCW         | m4  | +0.013 [−0.020, +0.050] | 0.075 | 44  | 0.83    | +0.158 [+0.00, +0.32] |
| mimic_full                  | m4  | +0.057 [+0.007, +0.110] | 0.100 | 41  | 0.78    | +0.202 [+0.04, +0.37] |
| NP-gate×BW_all_q50          | m4  | +0.003 [−0.023, +0.030] | 0.078 | 46  | 0.92    | +0.134 [+0.01, +0.26] |
| NP-gate×BW_all_q70          | m4  | +0.000 [−0.023, +0.020] | 0.115 | 48  | 0.95    | +0.063 [−0.03, +0.17] |
| proagate-all_q60_ablate     | m4  | +0.030 [−0.007, +0.067] | 0.097 | 38  | 0.73    | +0.112 [−0.05, +0.28] |
| proagate-all_q60_clamp_q30  | m4  | +0.043 [+0.003, +0.083] | 0.101 | 45  | 0.75    | +0.061 [−0.10, +0.23] |
| proagate-all_q60_otq_cal    | m4  | −0.003 [−0.030, +0.023] | 0.092 | 49  | 0.90    | +0.050 [−0.07, +0.18] |
| proagate-all_q75_ablate     | m4  | +0.017 [−0.013, +0.050] | 0.101 | 39  | 0.77    | +0.151 [−0.00, +0.32] |
| proagate-all_q75_clamp_q30  | m4  | +0.020 [−0.013, +0.057] | 0.083 | 45  | 0.81    | +0.074 [−0.08, +0.23] |
| proagate-all_q75_otq_cal    | m4  | +0.003 [−0.020, +0.030] | 0.088 | 46  | 0.94    | +0.070 [−0.03, +0.18] |
| prompt_hedge                | m4  | −0.003 [−0.063, +0.060] | 0.120 | 60  | 0.66    | +0.066 [−0.10, +0.24] |
| subspace-BW calibrated      | m4  | −0.010 [−0.050, +0.030] | 0.085 | 51  | 0.82    | +0.068 [−0.07, +0.23] |
| sweep_lda-allq50_ablate     | m4  | +0.013 [−0.020, +0.050] | 0.068 | 37  | 0.82    | +0.261 [+0.10, +0.42] |
| sweep_lda-allq50_clamp_q30  | m4  | +0.033 [−0.003, +0.073] | 0.113 | 45  | 0.82    | +0.107 [−0.05, +0.26] |
| sweep_lda-allq50_clamp_q50  | m4  | +0.023 [−0.007, +0.057] | 0.106 | 42  | 0.92    | +0.249 [+0.11, +0.39] |
| sweep_ocwcr-crq50_ablate    | m4  | +0.043 [−0.003, +0.093] | 0.082 | 28  | 0.79    | +0.350 [+0.19, +0.51] |
| sweep_ocwcr-crq50_clamp_q30 | m4  | +0.030 [−0.020, +0.080] | 0.080 | 44  | 0.79    | +0.177 [+0.01, +0.34] |
| sweep_ocwcr-crq50_clamp_q50 | m4  | +0.000 [−0.053, +0.053] | 0.089 | 42  | 0.83    | +0.293 [+0.13, +0.44] |
| baseline                    | m2  | —                       | 0.387 | 127 | —       | —                     |
| act_105                     | m2  | +0.017 [−0.010, +0.043] | 0.371 | 121 | 0.97    | +0.072 [+0.02, +0.14] |
| act_110                     | m2  | +0.000 [−0.030, +0.030] | 0.383 | 126 | 0.93    | +0.038 [−0.02, +0.11] |
| cast_gate-cr_q50_add-0.75   | m2  | −0.123 [−0.180, −0.067] | 0.246 | 95  | 0.56    | +0.029 [−0.09, +0.15] |
| ablation                    | m2  | +0.013 [−0.037, +0.063] | 0.349 | 120 | 0.86    | +0.116 [+0.02, +0.21] |
| additive −0.75              | m2  | −0.123 [−0.183, −0.057] | 0.126 | 74  | 0.44    | +0.146 [+0.03, +0.25] |
| clamp_q30                   | m2  | +0.000 [−0.053, +0.050] | 0.375 | 127 | 0.83    | +0.086 [−0.01, +0.18] |
| clamp_q50-think             | m2  | +0.037 [−0.010, +0.083] | 0.360 | 119 | 0.90    | +0.127 [+0.04, +0.21] |
| clamp_q50                   | m2  | +0.047 [+0.003, +0.093] | 0.333 | 112 | 0.90    | +0.180 [+0.09, +0.27] |
| clamp_q70                   | m2  | +0.030 [−0.013, +0.080] | 0.367 | 122 | 0.89    | +0.107 [+0.02, +0.19] |
| gated clamp_q50             | m2  | +0.020 [−0.023, +0.063] | 0.372 | 124 | 0.90    | +0.093 [+0.01, +0.18] |
| gated OT_q50                | m2  | +0.020 [−0.013, +0.053] | 0.367 | 124 | 0.95    | +0.062 [−0.00, +0.13] |
| gated clamp_q70             | m2  | +0.017 [−0.020, +0.053] | 0.372 | 123 | 0.92    | +0.080 [+0.01, +0.16] |
| gated OT_q70                | m2  | +0.020 [−0.007, +0.047] | 0.363 | 120 | 0.98    | +0.078 [+0.02, +0.14] |
| Gaussian-OT cal.            | m2  | +0.003 [−0.027, +0.037] | 0.386 | 125 | 0.95    | +0.049 [−0.01, +0.11] |
| otq_cal-think               | m2  | −0.020 [−0.050, +0.007] | 0.406 | 132 | 0.92    | +0.011 [−0.05, +0.08] |
| quantile-OT cal.            | m2  | +0.017 [−0.017, +0.053] | 0.374 | 122 | 0.93    | +0.062 [−0.01, +0.14] |
| quantile-OT non-OCW         | m2  | +0.000 [−0.033, +0.033] | 0.385 | 127 | 0.93    | +0.038 [−0.03, +0.10] |
| mimic_full                  | m2  | +0.033 [−0.007, +0.073] | 0.340 | 116 | 0.90    | +0.127 [+0.04, +0.21] |
| NP-gate×BW_all_q50          | m2  | +0.003 [−0.017, +0.023] | 0.382 | 125 | 0.98    | +0.023 [−0.02, +0.07] |
| NP-gate×BW_all_q70          | m2  | +0.007 [−0.010, +0.023] | 0.379 | 125 | 0.99    | +0.019 [−0.01, +0.06] |
| proagate-all_q60_ablate     | m2  | +0.010 [−0.023, +0.047] | 0.355 | 120 | 0.93    | +0.086 [+0.01, +0.16] |
| proagate-all_q60_clamp_q30  | m2  | +0.000 [−0.037, +0.033] | 0.382 | 126 | 0.91    | +0.036 [−0.04, +0.11] |
| proagate-all_q60_otq_cal    | m2  | +0.007 [−0.013, +0.027] | 0.386 | 126 | 0.98    | +0.015 [−0.03, +0.06] |
| proagate-all_q75_ablate     | m2  | +0.013 [−0.017, +0.047] | 0.364 | 120 | 0.95    | +0.080 [+0.02, +0.15] |
| proagate-all_q75_clamp_q30  | m2  | +0.013 [−0.017, +0.043] | 0.372 | 123 | 0.94    | +0.035 [−0.03, +0.10] |
| proagate-all_q75_otq_cal    | m2  | +0.003 [−0.020, +0.027] | 0.385 | 127 | 0.98    | +0.015 [−0.02, +0.06] |
| prompt_hedge                | m2  | +0.000 [−0.050, +0.050] | 0.372 | 126 | 0.83    | +0.078 [−0.02, +0.17] |
| subspace-BW calibrated      | m2  | +0.013 [−0.023, +0.050] | 0.374 | 122 | 0.93    | +0.070 [+0.00, +0.14] |
| sweep_lda-allq50_ablate     | m2  | +0.007 [−0.020, +0.033] | 0.365 | 124 | 0.95    | +0.041 [−0.02, +0.10] |
| sweep_lda-allq50_clamp_q30  | m2  | +0.010 [−0.017, +0.040] | 0.373 | 122 | 0.94    | +0.043 [−0.02, +0.11] |
| sweep_lda-allq50_clamp_q50  | m2  | +0.007 [−0.020, +0.033] | 0.372 | 123 | 0.95    | +0.047 [−0.01, +0.11] |
| sweep_ocwcr-crq50_ablate    | m2  | +0.020 [−0.027, +0.070] | 0.352 | 122 | 0.84    | +0.088 [−0.00, +0.18] |
| sweep_ocwcr-crq50_clamp_q30 | m2  | +0.033 [−0.010, +0.080] | 0.348 | 118 | 0.90    | +0.134 [+0.05, +0.22] |
| sweep_ocwcr-crq50_clamp_q50 | m2  | +0.033 [−0.010, +0.077] | 0.347 | 117 | 0.89    | +0.136 [+0.05, +0.22] |

Table 6. MMLU: all conditions, both readouts.

| condition                | r/o | $\Delta\text{acc}$ [CI]   | ECE   | OCW | cr_keep | Sel [CI]                |
|--------------------------|-----|---------------------------|-------|-----|---------|-------------------------|
| baseline                 | m4  | —                         | 0.092 | 37  | —       | —                       |
| additive $-0.75$         | m4  | $-0.123 [-0.187, -0.060]$ | 0.265 | 0   | 0.00    | $+0.000 [+0.00, +0.00]$ |
| clamp $q_{50}$           | m4  | $+0.037 [-0.013, +0.087]$ | 0.116 | 41  | 0.82    | $+0.197 [+0.03, +0.36]$ |
| gated clamp $q_{50}$     | m4  | $+0.027 [-0.010, +0.063]$ | 0.124 | 41  | 0.89    | $+0.137 [-0.01, +0.29]$ |
| quantile-OT non-OCW      | m4  | $-0.027 [-0.060, +0.007]$ | 0.073 | 36  | 0.91    | $+0.095 [-0.03, +0.24]$ |
| NP-gate×BW all $q_{50}$  | m4  | $-0.017 [-0.040, +0.007]$ | 0.090 | 38  | 0.94    | $+0.052 [-0.05, +0.17]$ |
| subspace-BW calibrated   | m4  | $-0.007 [-0.043, +0.030]$ | 0.127 | 40  | 0.91    | $+0.048 [-0.07, +0.18]$ |
| sweep_ocwcr-crq50_ablate | m4  | $-0.003 [-0.040, +0.033]$ | 0.118 | 38  | 0.84    | $+0.222 [+0.05, +0.39]$ |
| baseline                 | m2  | —                         | 0.229 | 75  | —       | —                       |
| additive $-0.75$         | m2  | $-0.200 [-0.260, -0.140]$ | 0.126 | 90  | 0.56    | $+0.016 [-0.11, +0.15]$ |
| clamp $q_{50}$           | m2  | $-0.020 [-0.060, +0.020]$ | 0.247 | 81  | 0.90    | $+0.128 [+0.03, +0.23]$ |
| gated clamp $q_{50}$     | m2  | $-0.027 [-0.063, +0.003]$ | 0.255 | 83  | 0.92    | $+0.057 [-0.02, +0.14]$ |
| quantile-OT non-OCW      | m2  | $+0.003 [-0.017, +0.023]$ | 0.223 | 71  | 0.99    | $+0.080 [+0.02, +0.15]$ |
| NP-gate×BW all $q_{50}$  | m2  | $-0.003 [-0.020, +0.013]$ | 0.231 | 74  | 0.99    | $+0.040 [-0.01, +0.10]$ |
| subspace-BW calibrated   | m2  | $-0.017 [-0.040, +0.003]$ | 0.250 | 80  | 0.97    | $+0.022 [-0.03, +0.08]$ |
| sweep_ocwcr-crq50_ablate | m2  | $-0.007 [-0.040, +0.023]$ | 0.234 | 77  | 0.94    | $+0.097 [+0.01, +0.19]$ |

Table 7. ARC-Challenge (transfer): all conditions, both read-outs.

| condition                | r/o | $\Delta\text{acc}$ [CI]   | ECE   | OCW | cr_keep | Sel [CI]                |
|--------------------------|-----|---------------------------|-------|-----|---------|-------------------------|
| baseline                 | m4  | —                         | 0.280 | 6   | —       | —                       |
| additive $-0.75$         | m4  | $-0.257 [-0.330, -0.187]$ | 0.069 | 0   | 0.00    | $+0.000 [+0.00, +0.00]$ |
| clamp $q_{50}$           | m4  | $+0.007 [-0.047, +0.060]$ | 0.286 | 6   | 0.42    | $-0.083 [-0.64, +0.45]$ |
| gated clamp $q_{50}$     | m4  | $+0.020 [-0.027, +0.070]$ | 0.295 | 7   | 0.50    | $+0.167 [-0.41, +0.67]$ |
| quantile-OT non-OCW      | m4  | $+0.010 [-0.030, +0.050]$ | 0.278 | 6   | 0.58    | $+0.083 [-0.47, +0.62]$ |
| NP-gate×BW all $q_{50}$  | m4  | $+0.000 [+0.000, +0.000]$ | 0.280 | 6   | 1.00    | $+0.000 [+0.00, +0.00]$ |
| NP-gate×BW self $q_{70}$ | m4  | $+0.007 [-0.023, +0.037]$ | 0.272 | 6   | 0.83    | $-0.167 [-0.43, +0.00]$ |
| subspace-BW calibrated   | m4  | $-0.007 [-0.050, +0.033]$ | 0.266 | 7   | 0.67    | $-0.167 [-0.57, +0.30]$ |
| baseline                 | m2  | —                         | 0.201 | 81  | —       | —                       |
| additive $-0.75$         | m2  | $-0.323 [-0.393, -0.257]$ | 0.241 | 107 | 0.28    | $-0.132 [-0.26, -0.00]$ |
| clamp $q_{50}$           | m2  | $-0.027 [-0.070, +0.013]$ | 0.221 | 80  | 0.87    | $+0.179 [+0.06, +0.29]$ |
| gated clamp $q_{50}$     | m2  | $-0.010 [-0.050, +0.027]$ | 0.212 | 83  | 0.91    | $+0.140 [+0.03, +0.24]$ |
| quantile-OT non-OCW      | m2  | $+0.007 [-0.023, +0.037]$ | 0.181 | 76  | 0.94    | $+0.108 [+0.02, +0.21]$ |
| NP-gate×BW all $q_{50}$  | m2  | $+0.000 [+0.000, +0.000]$ | 0.201 | 81  | 1.00    | $+0.000 [+0.00, +0.00]$ |
| NP-gate×BW self $q_{70}$ | m2  | $+0.003 [-0.020, +0.027]$ | 0.193 | 78  | 0.96    | $+0.108 [+0.03, +0.19]$ |
| subspace-BW calibrated   | m2  | $+0.003 [-0.033, +0.040]$ | 0.202 | 75  | 0.92    | $+0.204 [+0.10, +0.31]$ |

Table 8. GSM8K-MCQ (transfer): all conditions, both read-outs. Note the tiny OCW/CR poles (6/12 of 300 at baseline); the model is underconfident on this domain, selectivity CIs are wide, and transport maps correctly act near-identity.

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вмешательство для управления рассуждениями языковых моделей**  
стиринг активаций and оптимальный транспорт and калибровка and  
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